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AI for Bharat Hackathon

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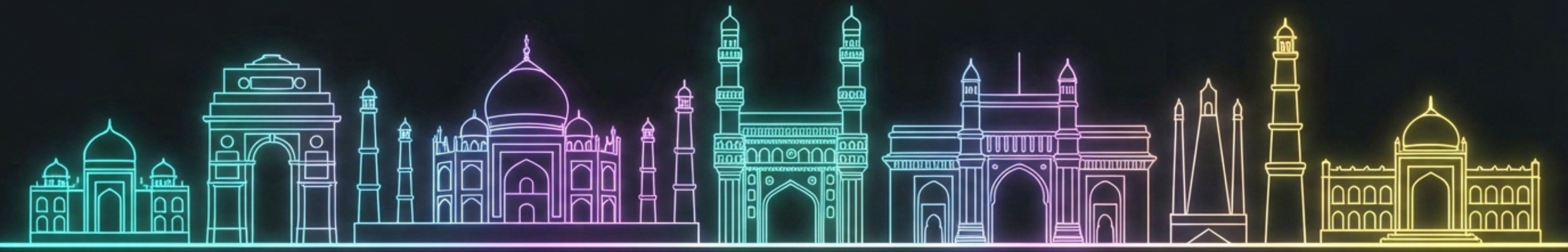

Team Name: Choco Warriors

Team Leader: Shashank Hara Sridhar

Problem Statement: Design an AI solution that improves efficiency, understanding, or support within healthcare or life-sciences ecosystems.

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Healthcare is becoming a luxury, but the hidden cost is information asymmetry.

Financial Opacity

Patients cannot access true costs or verified doctor fees before treatment, leading to financial strain.

The Trust Deficit

Existing platforms rely on subjective, unverified text prone to fake 5-star ratings.

Insurance Confusion

Lack of clarity on accepted insurance policies and actual claim success rates.

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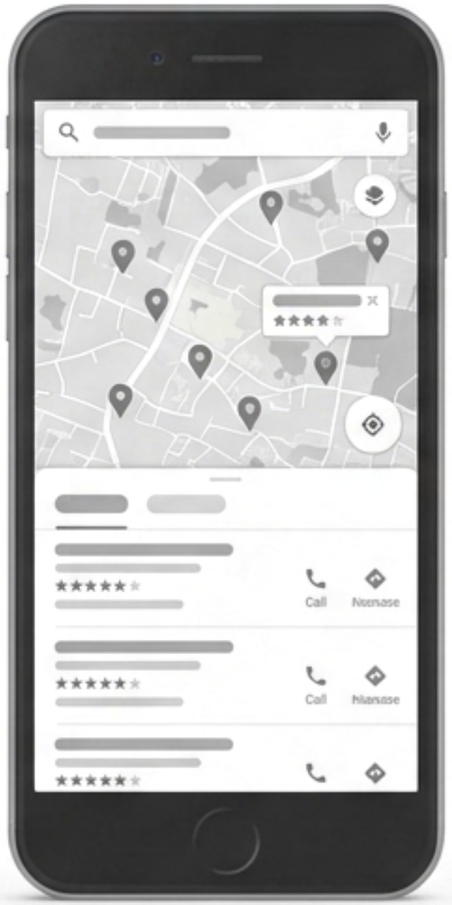
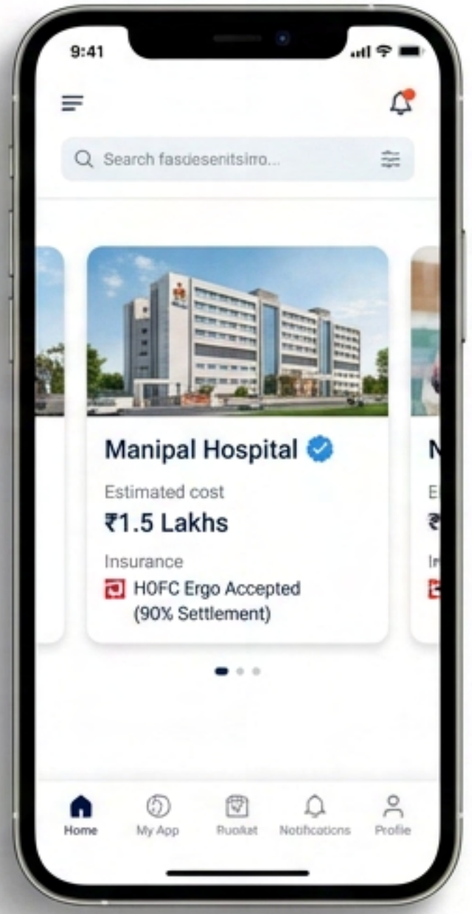
Patients don’t need a directory; they need a decision assistant.

The Old Way (Keyword Search)



The New Way (Intent-Based Swasthya Saathi)



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How Different Is It From Other Existing Ideas?

Existing Ideas

Unverified Star Ratings & Simple Keyword Searches

Rely on subjective reviews and basic text matching, often leading to misleading information.

This Solution

Full Medical Intent Understanding

Moves beyond simple keyword searches to understand the user's specific medical needs.

Verified Evidence over Star Ratings

Prioritizes data from authentic documents over unverified and often misleading star ratings.

Verified Cost & Outcome Signals

Extracts actual pricing and treatment outcomes directly from uploaded medical receipts and documents.

Integrated Insurance Matrix

Directly factors insurance acceptance and claim history into the decision-making process.

Transparent Hospital Reasoning & Data Fallback Logic

Provides a clear explanation for every recommendation based on cost, quality, and budget, using data fallback logic for continuous support.

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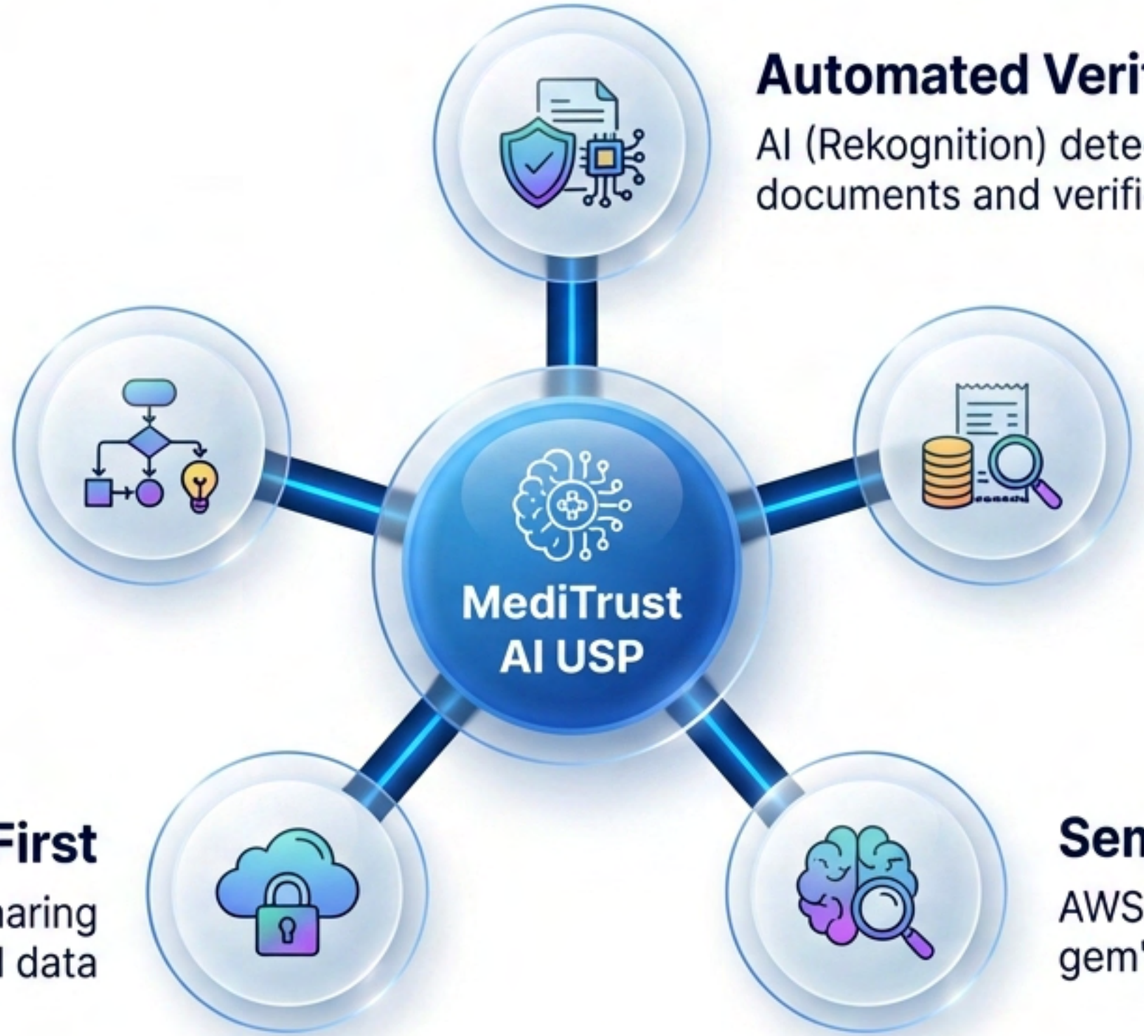
USP of the proposed solution



Automated Verification

AI (Rekognition) detects fake documents and verifies authenticity

Explanation Engine

Human-readable reasons for every hospital recommendation

Financial Transparency

Real-world pricing extracted from uploaded receipts

Privacy-First

Secure, anonymized sharing of medical data

Semantic Search

AWS OpenSearch finding 'hidden gem' doctors based on experience

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List of features offered

Natural Language Search

Ask complex medical questions like a consultant.

Verified Reviews

Marked 'Verified' only after AI validates proof of visit.

Cost Estimator

Real-time visibility into amounts paid by other patients.

Insurance Compatibility

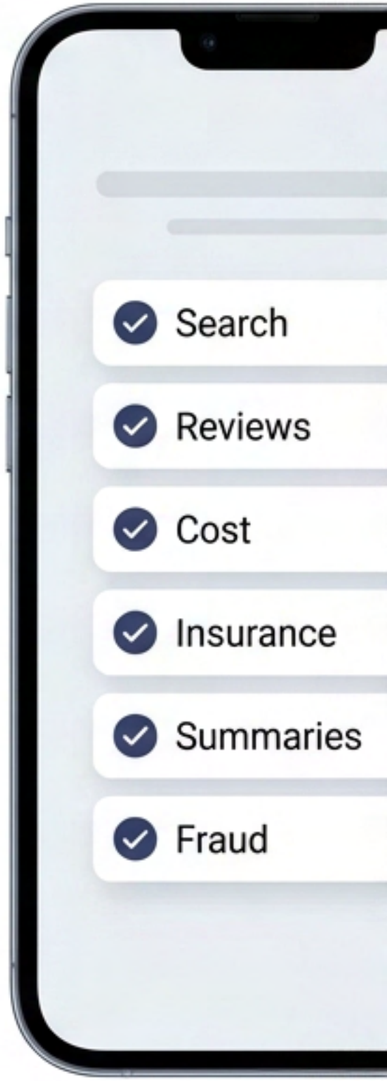
Filter by successful claim histories.

Smart Summaries

AI-generated pros and cons for every profile.

Fraud Detection

Automated flagging of manipulated images.



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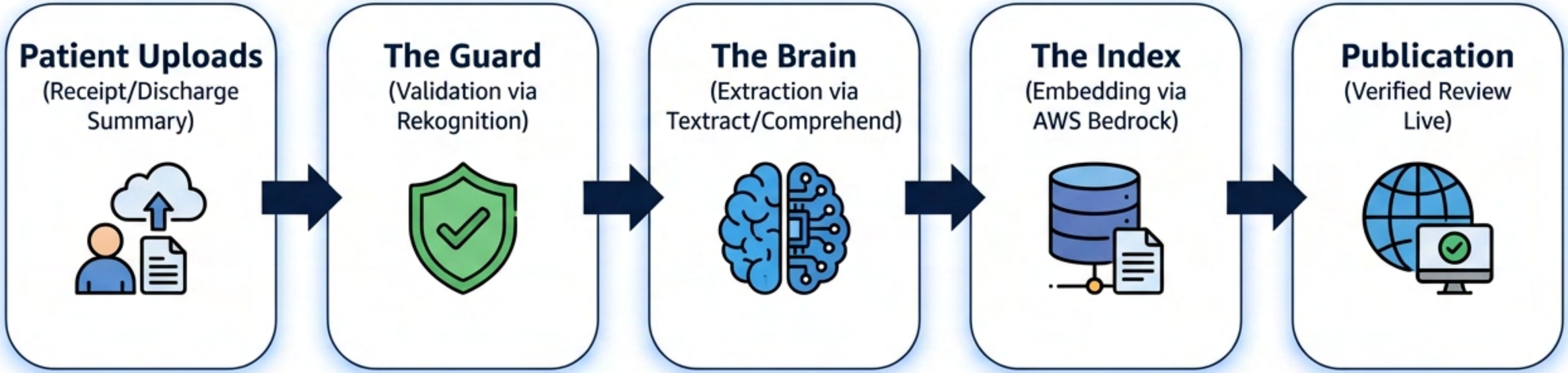
Process flow diagram



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AWS Healthcare Architecture: Scalable Document Processing & Medical Insights

Access and Authentication

Secure User Entry Point

Amazon CloudFront

AWS Amplify

Users interact with the system via Amazon CloudFront and AWS Amplify, providing a responsive and secure front-end interface.

Identity Management

Amazon Cognito

JWT

Amazon Cognito manages JWT tokens to ensure authenticated and authorized access to backend resources.

Request Routing

Amazon API Gateway

Amazon API Gateway acts as the central hub, routing requests to specific microservices like Hospital, Doctor, and insurance services.

Hospital Lambda

Doctor Lambda

Insurance Lambda

Path

Path

The Core Workflow (AWS Step Functions)

AWS Step Functions

HyperVerge

Document Validation

The "Document Validation Service" utilizes HyperVerge (via AWS Marketplace) for identity and document authenticity verification.

Amazon Rekognition

Amazon Textract

Medical Text Extraction

Images and documents are processed by Amazon Rekognition and Amazon Textract to pull raw text and visual data.

Amazon Comprehend Medical

Medical Entity Recognition

Amazon Comprehend Medical analyzes extracted text to identify and categonze medical terms and patient data.

Amazon Bedrock

Embedding and Indexing

Amazon Bedrock generates embeddings for the processed data, which are then indesed for advanced search capabilities.

Monitoring and Support

Logs

Logs

Amazon CloudWatch

Centralized Observability

Amazon CloudWatch aggregates logs from all Lambda functions and the API Gateway for real-time monitoring and troubleshooting.

Customer Support Integration

Customer Service Lambda

A dedicated Customer Service Lambda manages customer-specific data and interactions stored within the S3 environment.

Data Storage and Search

Path

Path

Structured Data Management

Amazon DynamoDB

Amazon DynamoDB stores high-velocity CRUD data for hospitals, doctors, insurance providers, and reviews.

Object Storage

Amazon S3 (Standard)

Amazon S3 (Standard) serves as the primary repository for storing original documents and customer-related files.

Intelligent Vector Search

Search Lambda

Amazon Bedrock

Amazon OpenSearch Service

A dedicated "Search Lambda" coordinates between Amazon Bedrock and Amazon OpenSearch Service to perform complex vector searches on indexed medical data.

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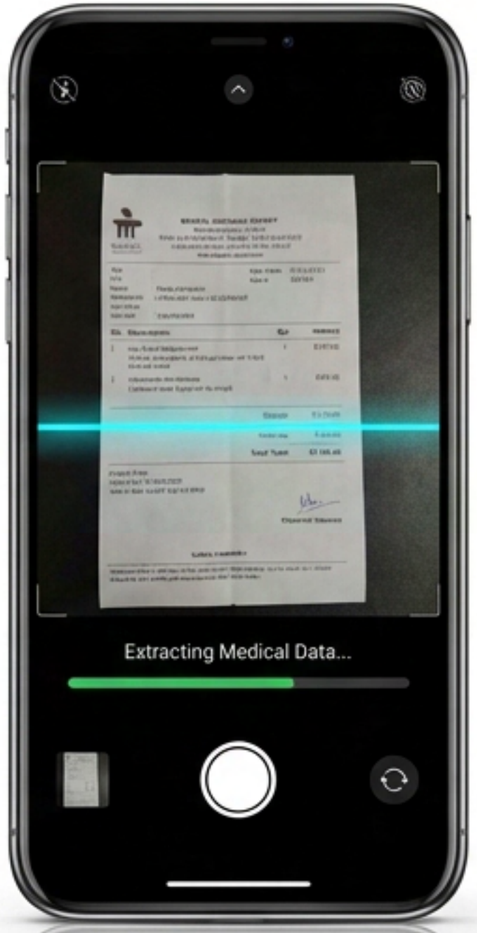
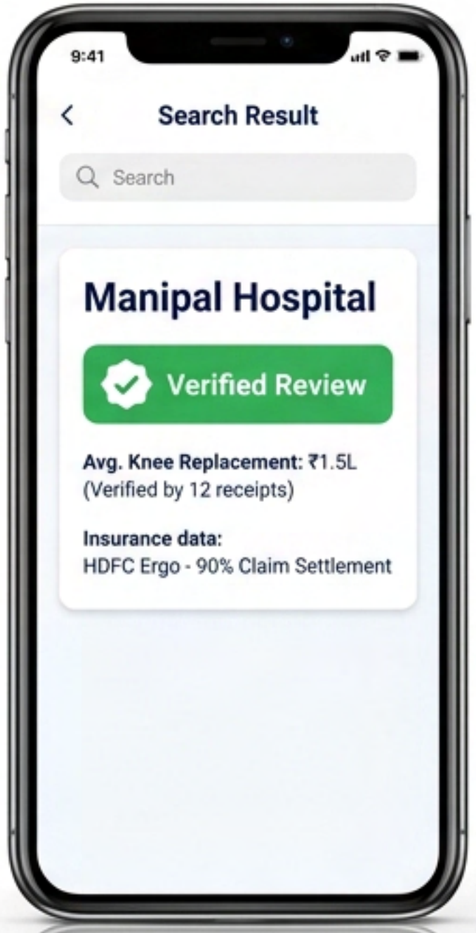
Wireframes / Mockups





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Technologies to be used in the solution

Frontend	React	AWS Amplify	Amazon CloudFront
Compute & Orchestration	AWS Lambda	AWS Step Functions	
AI & ML	Amazon Bedrock (Titan/Claude)	Amazon Rekognition	Amazon Textract	Amazon Comprehend Medical
Data & Storage	Amazon DynamoDB	Amazon S3	Amazon OpenSearch
Auth	Amazon Cognito		

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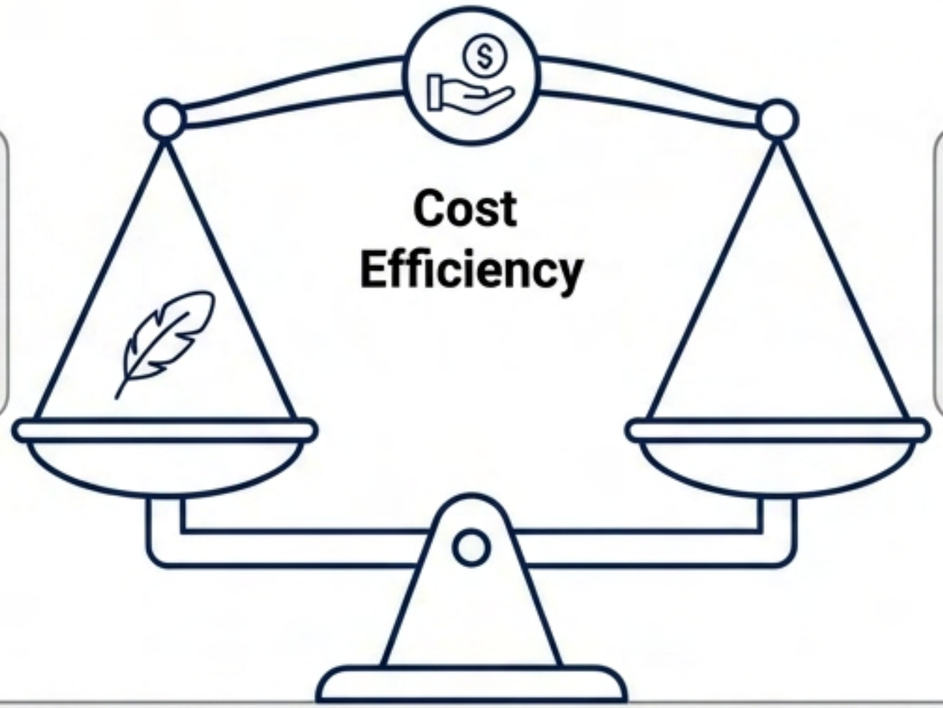
Estimated implementation cost

Serverless Efficiency

- Pay-per-use (Lambda/DynamoDB) means near-zero cost when idle.



Operational Savings

- No upfront infrastructure investment.
- No server management.

Estimate: Initial prototype runs primarily within AWS Free Tier. Production costs scale linearly with active users, ensuring business viability.

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Thank You

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